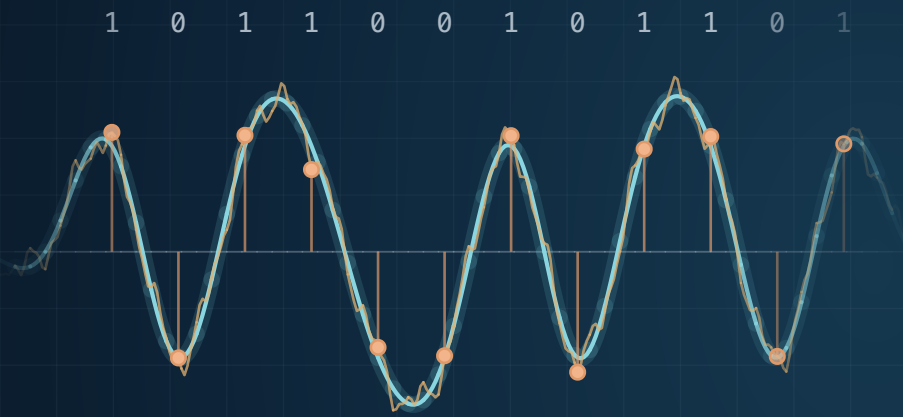




SAMPLING · DETECTION · MODULATION · CODING

# Digital Communications

*Formula and Notation Reference*



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CONVENTIONS · FORMULAS · NOTATION

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*The conventions used throughout the course, every formula it establishes, and every symbol it defines. Nothing here is derived; the derivations are in the lecture notes.*

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## How to read it

Part 1 states the conventions, the two expressions every error probability comes from, and the transform pair Chapters 1 and 2 use. Part 2 is the summary of formulas, in the order the course establishes them. Part 3 defines every symbol. Nothing here is derived: where a result needs an argument, the argument is in the lecture notes chapter named beside it.

## PART

## 1

# Conventions

*These hold everywhere in the course, without local variation. Half the factor-of-two errors in this subject come from mixing two of them in one line, so where a result depends on one it is restated at that point rather than assumed.*

| CONVENTION       | STATEMENT                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy and power | Normalised: the resistance is taken as $1 \Omega$ , so instantaneous power is $ x(t) ^2$ and energy is its integral. Signal energy is $E$ , energy per bit $E_b$ , energy per symbol $E_s$ , and $E_s = (\log_2 M) E_b$ . |
| Noise            | White and Gaussian with <b>two-sided</b> power spectral density $S_n(f) = N_0/2$ . Projected onto an orthonormal basis it gives independent components, each $\mathcal{N}(0, N_0/2)$ .                                    |
| The $Q$ function | $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ . It is the tail of a standard normal, so $Q(0) = \frac{1}{2}$ and $Q(-x) = 1 - Q(x)$ .                          |
| sinc             | Normalised: $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ , with zeros at every non-zero integer. Stated again wherever it is used.                                                                                      |
| Logarithms       | log without a base means base two, and every entropy and every bit count is in bits. Decibels are $10 \log_{10}$ of a power or energy ratio.                                                                              |
| Signals          | $s_i(t)$ is transmitted, $r(t)$ is received, $\{\psi_k(t)\}$ is the orthonormal basis, and boldface $\mathbf{s}_i$ is the point in signal space.                                                                          |
| Priors           | Symbols are equally likely unless a problem says otherwise, in which case the MAP form is used and the prior term written out.                                                                                            |

## THE TWO EXPRESSIONS THE WHOLE COURSE RESTS ON

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

The first is exact for two signal points and is where every error probability in Chapters 2, 4 and 5 comes from. The second is its nearest-neighbour extension to  $M$  points, and the 2 in the denominator is the noise variance per dimension being  $N_0/2$ .

## THE FOURIER TRANSFORM PAIR, AS CHAPTERS 1 AND 2 USE IT

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

Written in hertz rather than in rad/s, so no factor of  $2\pi$  appears in front of either integral. Sampling and the Nyquist criterion are both stated in this form.

## PART

## 2

# Summary of formulas

Everything the course establishes, in the order it establishes it. Nothing here is derived — where a result needs an argument, the argument is in the chapter named beside it.

## A.1 From an analog signal to bits — Chapter 1

### SAMPLING AND RECONSTRUCTION

$f_s > 2W$  (the sampling theorem,  $W$  the highest frequency present)

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Below the Nyquist rate the replicas of the spectrum overlap and the overlap cannot be undone.  $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$  throughout this course.

### UNIFORM QUANTIZATION

$$\Delta = \frac{2V}{L} = \frac{2V}{2^n}, \quad |e| \leq \frac{\Delta}{2}$$

$$\sigma_q^2 = \frac{\Delta^2}{12}, \quad \operatorname{SQNR}|_{\text{dB}} = 6.02n + 1.76$$

The 1.76 assumes a full-scale sinusoid. The formula is an approximation at small  $n$ . At three bits it gives 19.82 dB where the measured value is 19.09 dB, and the gap halves with each extra bit.

Companding and the PCM bit rate.

| RESULT     | STATEMENT                                                                                                  | CHAPTER |
|------------|------------------------------------------------------------------------------------------------------------|---------|
| Companding | $\mu$ -law and A-law compress before quantizing so that the signal-to-noise ratio is flat across the range | 1       |
| PCM rate   | $R = n f_s$ bits a second for $n$ bits a sample                                                            | 1       |

## A.2 Baseband transmission — Chapter 2

### THE MATCHED FILTER

$$h(t) = s(T - t), \quad \left( \frac{S}{N} \right)_{\max} = \frac{2E}{N_0}$$

The matched filter maximizes the signal-to-noise ratio at the sampling instant. The maximum depends on pulse energy, not pulse shape.

### BINARY ERROR PROBABILITY AT BASEBAND

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ (antipodal),} \quad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \text{ (on-off, orthogonal)}$$

### NYQUIST CRITERION FOR ZERO INTERSYMBOL INTERFERENCE

$$\sum_k P\left(f + \frac{k}{T}\right) = T \quad \text{for all } f$$

$$B = \frac{1 + \alpha}{2T}, \quad 0 \leq \alpha \leq 1$$

The raised cosine meets the criterion for every roll-off  $\alpha$ . At  $\alpha = 0$  the bandwidth is the Nyquist minimum  $1/2T$  and the pulse decays slowly. At  $\alpha = 1$  it is twice that and the pulse decays quickly.

## A.3 Signal space — Chapter 3

### A SIGNAL AS A POINT

$$s_i(t) = \sum_{k=1}^N s_{ik} \psi_k(t), \quad s_{ik} = \int_0^T s_i(t) \psi_k(t) dt$$

$$E_i = \|\mathbf{s}_i\|^2 = \sum_{k=1}^N s_{ik}^2, \quad d_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|$$

The basis  $\{\psi_k\}$  is orthonormal and comes from the Gram–Schmidt procedure applied to the signal set. Energy becomes a squared length and difference becomes a distance, which is the move the rest of the course rests on.

## A.4 The optimal receiver — Chapter 4

### THE DECISION RULE

$$\hat{s} = \arg \min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad (\text{MAP})$$

$$\hat{s} = \arg \min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad (\text{ML, equal priors})$$

With equal priors the rule is minimum distance. The equivalent correlation form is to maximise  $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$ .

### ERROR PROBABILITY

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q \left( \sqrt{\frac{d_{kj}^2}{2N_0}} \right)$$

$$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q \left( \sqrt{\frac{d_{kj}^2}{2N_0}} \right), \quad P_e \approx N_{\min} Q \left( \sqrt{\frac{d_{\min}^2}{2N_0}} \right)$$

The first result is exact for two points. The second is the union bound. The third is its nearest-neighbor form. Here,  $N_{\min}$  is the average number of nearest neighbors and can be noninteger. The loose bound needs only  $M$  and  $d_{\min}$ .

## A.5 Digital modulation — Chapter 5

Minimum squared distance and number of nearest neighbours for each scheme.

| SCHEME            | $d_{\min}^2$          | $N_{\min}$                      |
|-------------------|-----------------------|---------------------------------|
| BPSK              | $4E_b$                | 1                               |
| BFSK, BASK        | $2E_b$                | 1                               |
| $M$ -PSK          | $4E_s \sin^2(\pi/M)$  | 2                               |
| $M$ -PAM          | $12E_s/(M^2 - 1)$     | $2(M - 1)/M$                    |
| $M$ -QAM (square) | $6E_s/(M - 1)$        | 3 at $M = 16$ , 3.5 at $M = 64$ |
| $M$ -FSK          | $2E_s$ for every pair | $M - 1$                         |

### READING THE TABLE

$$P_e \approx N_{\min} Q \left( \sqrt{\frac{d_{\min}^2}{2N_0}} \right), \quad E_s = (\log_2 M) E_b$$

Convert once between symbol and bit energy. BFSK and BASK need 3.01 dB more than BPSK for the same error probability. Doubling  $M$  costs approximately 6 dB in PAM. Moving from 4-PSK to 8-PSK costs 5.33 dB. At the same energy, 16-QAM gains 9.29 dB over 16-PAM.

## A.6 Information theory — Chapter 6

### INFORMATION AND ENTROPY

$$I(s_k) = -\log_2 p_k, \quad H(S) = -\sum_{k=1}^K p_k \log_2 p_k$$

$$0 \leq H(S) \leq \log_2 K, \quad H(S^n) = n H(S)$$

The upper bound is reached when all  $K$  symbols are equally likely. The extension result holds because the source is memoryless.

### SOURCE CODING

$$\bar{L} = \sum_{k=1}^K p_k l_k, \quad \eta = \frac{H(S)}{\bar{L}} \leq 1$$

$$\sum_{k=1}^K 2^{-l_k} \leq 1, \quad H(S) \leq \bar{L} < H(S) + 1$$

The Kraft inequality is necessary but not sufficient for a prefix code. Over the  $n$ -th extension the upper bound becomes  $H(S) + 1/n$ , so  $L_n/n \rightarrow H(S)$ .

### HUFFMAN CODES

$$\sigma^2 = \sum_{k=1}^K p_k (l_k - \bar{L})^2$$

Huffman is optimal in average length and is not unique. Placing a merged symbol as high as possible on a tie gives the minimum-variance code among the optimal ones.



## PART

## 3

# Notation

---

Every symbol the course defines, with the chapter that defines it. A symbol is never reused for a second meaning.

$f_s, T_s$

Sampling rate in hertz and the interval between samples,  $T_s = 1/f_s$ .

$W$

The highest frequency present in a message. The sampling theorem asks for  $f_s > 2W$ .

$\text{sinc}(x)$

Normalised:  $\sin(\pi x)/(\pi x)$ , with zeros at every non-zero integer.

$\Delta, L, n$

Quantizer step, number of levels and bits a sample, with  $L = 2^n$  and  $\Delta = 2V/L$  over a range  $\pm V$ .

$\sigma_q^2$

Quantization noise power,  $\Delta^2/12$  for a uniform quantizer.

$T, R$

Symbol interval and the rate that follows from it. A bit interval is written  $T_b$  and a bit rate  $R_b$ .

$p(t), h(t)$

The transmitted pulse and the impulse response of the receive filter. The matched filter is  $h(t) = p(T - t)$ .

$\alpha$

Roll-off of a raised-cosine pulse, from 0 to 1. Bandwidth is  $(1 + \alpha)/2T$ .

$\psi_k(t)$

The  $k$ -th orthonormal basis function, from the Gram–Schmidt procedure applied to the signal set.

$s_i(t), \mathbf{s}_i$

The  $i$ -th transmitted waveform, and the point in signal space whose coordinates are its projections onto the basis.

$E, E_i$

Signal energy, normalised so that no resistance appears:  $E_i = \|\mathbf{s}_i\|^2$ .

$d_{ij}, d_{\min}$

Distance between two signal points, and the smallest such distance in a constellation.

$r(t), \mathbf{r}$

The received waveform and the observation vector it produces,  $\mathbf{r} = \mathbf{s}_i + \mathbf{n}$ .

$N_0$

Noise power spectral density parameter. The two-sided density is  $N_0/2$  watts per hertz, and each noise component is  $\mathcal{N}(0, N_0/2)$ .

$Q(x)$ 

The Gaussian tail: the probability that a standard normal variable exceeds  $x$ . Equal to  $\frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ .

 $M, N$ 

The number of signals in the set, and the number of basis functions they need.  $M$  signals carry  $\log_2 M$  bits a symbol.

 $N_{\min}$ 

The average number of signal points at the minimum distance. An average, so not usually a whole number.

 $P_b, P_e, P_s$ 

Bit error probability, error probability in general, and symbol error probability.

 $E_b, E_s$ 

Energy per information bit and energy per transmitted symbol, with  $E_s = (\log_2 M) E_b$ .

 $f_c$ 

Carrier frequency. Every scheme in Chapter 5 changes the amplitude, the phase or the frequency of  $\cos(2\pi f_c t)$ .

 $S, K, p_k$ 

The source alphabet, how many symbols it has, and the probability of the  $k$ -th.

 $I(s_k)$ 

Self-information of one symbol,  $-\log_2 p_k$  bits.

 $H(S)$ 

Entropy: the average information a symbol carries, in bits a symbol. Bounded by 0 and  $\log_2 K$ .

 $\bar{L}, l_k$ 

Average codeword length and the length of the codeword for symbol  $k$ .

 $\eta$ 

Coding efficiency,  $H(S)/\bar{L}$ . Never above one.

 $\sigma^2$ 

Variance of the codeword length about  $\bar{L}$ . Separates two Huffman codes that share an average.

 $p(y_k | x_j)$ 

Transition probability of a discrete memoryless channel. Collected into the channel matrix, whose every row sums to one.

 $H(X | Y)$ 

Conditional entropy: the uncertainty still remaining about the input after the output has been seen.

 $I(X; Y)$ 

Mutual information,  $H(X) - H(X | Y)$ : what the channel actually delivered, in bits per use. Symmetric and never negative.

 $C$ 

Channel capacity, the largest  $I(X; Y)$  over all input distributions. For the binary symmetric channel,  $1 - H(p)$ .

 $H(p)$ 

The binary entropy function,  $-p \log_2 p - (1 - p) \log_2 (1 - p)$ . It measures a two-symbol source and also the noise a binary symmetric channel adds.

**Digital Communications · Formula and Notation Reference**

| VERSION | DATE              | CHANGES        |
|---------|-------------------|----------------|
| v0      | 24 September 2026 | First edition. |

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